

Nightingale Cloud

Shared Longitudinal Care Note — Technical Brief

Synthetic-data-only 72-hour prototype

A communication and trust layer that turns fragmented longitudinal clinical notes into an actionable, provenance-linked shared record.

1. Problem & Design Principles

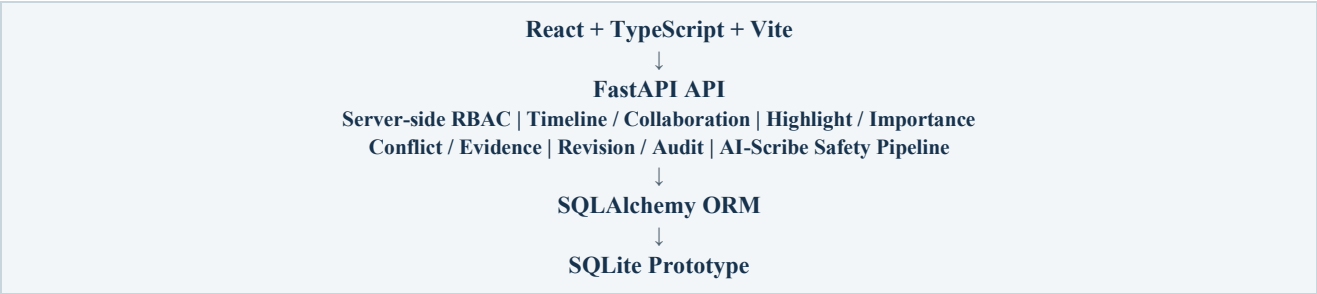
Electronic health records store structured observations and encounter documents, but they often fail to express the patient's evolving story. Free-text notes are fragmented by visit, author, and workflow; important risks compete with old details; unresolved actions are easy to lose; and AI summaries add value only when clinicians can verify their evidence.

Nightingale Cloud centers one shared Care Note per patient. It combines a longitudinal timeline with a clinician-facing Glance View, inline collaboration, version history, and AI-scribed entries. It is a focused 72-hour prototype, not a production EHR.

- **Glanceability:** surface the three to five most actionable facts and open actions first
- **Collaboration:** let staff and clinicians contribute without overwriting each other's sections
- **Provenance:** trace every highlighted claim to an exact timeline entry and source span
- **Deterministic safety:** use inspectable authority, conflict, ranking-floor, and evidence rules
- **Abstention:** withhold or require review when privacy or evidence checks fail

2. Architecture

FastAPI resolves a development identity, loads the patient, enforces reusable server-side authorization, and delegates to focused services. SQLAlchemy models and Pydantic schemas remain separate. The React client is a typed REST consumer; its role switcher is demo simulation, not the security boundary.



Synthetic Transcript → PHI Redaction → Redaction Validation	
PASS	FAIL
Provider called	WITHHELD - provider not called
System-authored AI timeline entry	No AI entry

3. Data Model

TimelineEntry is the canonical longitudinal unit. Immutable EntryVersion snapshots support history and revert. Updates carry an expected version: stale same-entry edits fail with HTTP 409, while unrelated entries remain independently editable. Conflict records retain both sources; AI-derived instructions reuse timeline entries with approval metadata rather than a parallel content store.

Longitudinal core	Trust links	Governance
Patient → TimelineEntry TimelineEntry → EntryVersion TimelineEntry → Comment / TaskAssignment	TimelineEntry → Highlight Highlight → provenance pointer Pointer → entry + exact source span ConflictRecord → both source entries	AI instruction → source AI entry → approval metadata AuditLog → metadata-only events Clinic → adaptive feedback counters

4. Glance, Prioritization & Trust

The Glance View returns up to five non-rejected highlights. The base score combines explicit risk, recency, unresolved work, clinical entity type, and clinician confirmation. Clinic-scoped accept/reject counters add a bounded entity and entry-type adjustment. Feedback changes ranking only; it cannot change provenance, authority, approval state, or Evidence Confidence.

$$\text{base score} + \text{learned adjustment} = \text{adjusted score}$$

$$\text{final score} = \max(\text{adjusted score}, \text{clinical safety floor})$$

Safety floors apply after learning: allergy remains at least HIGH/50; unresolved medication dosage conflict HIGH/65; recent or unresolved medication change MODERATE/35; and unresolved clinically relevant follow-up MODERATE/50. Repeated negative feedback cannot demote a critical class below its floor.

Evidence Confidence is not LLM self-reported confidence. It is a deterministic state derived from verifiable evidence.

Level	Meaning	System behavior
HIGH	Verified source/span plus deterministic evidence or clinician confirmation	Surface normally
MEDIUM	Verified provenance without deterministic structured confirmation	Surface with evidence context
LOW	Source exists but deterministic evidence is inconsistent	Needs review
ABSTAIN	Broken provenance, invalid span, or unresolved contradiction	Do not present as trusted

What is it? A deterministic evidence state. **How do we know if it is wrong?** Validate provenance, source spans, structured facts, and conflicts. **What happens when it is wrong?** Abstain or require clinician review.

5. AI Safety, Conflicts & Patient Safety

AI scribe	Patient-facing gate	After approval
Transcript → Redact PHI → Validate	AI-derived instruction → Draft	Content edited or reverted
PASS → Provider → AI entry	Clinician review → Approved → Patient visible	Approval invalidated → Draft
FAIL → WITHHELD → no provider / entry	Rejected → remains internal	Requires clinician re-approval

Every provider call follows redaction and validation. Names, Singapore-style identity numbers, and phones are redacted; validation checks remaining PHI patterns, protected clinical terminology, medication dosages, and output integrity. Failure returns WITHHELD, invokes no provider, logs no raw transcript, and creates no AI entry.

Conflict extraction is limited to the synthetic medication/dosage, allergy, and follow-up vocabulary. It compares clinician with AI/patient evidence, clinician with staff, and staff with staff. The authority hierarchy is clinician, then staff/nurse, then AI/patient-derived evidence. Higher-authority evidence takes precedence without deleting the lower source. Equal-authority human contradictions stay open and require clinician review; the system does not invent truth.

There is no direct AI-to-patient path. AI-derived instructions begin as Draft and remain invisible until clinician approval. Rejected items stay internal. Editing or reverting approved content invalidates approval, returns it to Draft, and requires re-approval while prior versions remain available.

6. RBAC, Privacy & Audit

Authorization and clinic isolation are server-side. Patients retrieve only approved patient-facing instructions and cannot access raw AI notes or internal comments, conflicts, tasks, versions, or audit data. Staff manage staff notes within their clinic but cannot overwrite clinician notes or approve instructions. Clinicians manage clinician-owned notes, review AI/staff context, decide highlights and conflicts, and approve patient instructions. Admins have clinic-scoped oversight without implicit approval authority.

The prototype uses synthetic data only. AuditLog stores actor, role, action, entity identifiers, timestamps, and minimal status transitions; clinical, comment, highlight, instruction, and transcript text is excluded. Unauthorized and no-op operations do not create successful-action events. Development headers are identity injection, not production authentication. TLS and encryption at rest are deployment controls, not claims about this local build.

7. Performance & Verification

Warm-ups	20
Measured requests	200
Median	5.308 ms
P95	6.540 ms
Minimum	4.616 ms
Maximum	17.024 ms

Environment: Windows 11, Python 3.12.13, FastAPI TestClient, and in-memory SQLite.

This is an in-process backend warm-path approximation and does not include browser rendering, network latency, deployed infrastructure, production data volume, or concurrency.

Required micro-tests	18 passed
Complete backend suite	89 passed
Frontend TypeScript	PASS
Frontend production build	PASS
External LLM calls during verification	None

8. Trade-offs & Next Steps

Deliberate 72-hour scope choices include SQLite rather than a production database; SQLAlchemy create_all rather than migration infrastructure; development identity headers rather than production authentication; a deterministic mock AI provider; synthetic/demo clinical extraction vocabulary; no raw AI transcript viewer; data decay as a reversible policy preview rather than physical cold storage; and no ambient voice capture.

These choices prioritize a working longitudinal workflow, provenance, clinical authority, deterministic safety, evaluation, and clarity. Production next steps would begin with authenticated identity, managed migrations and storage, clinically validated extraction, direct source-review tooling, deployed performance testing, and operational security controls.